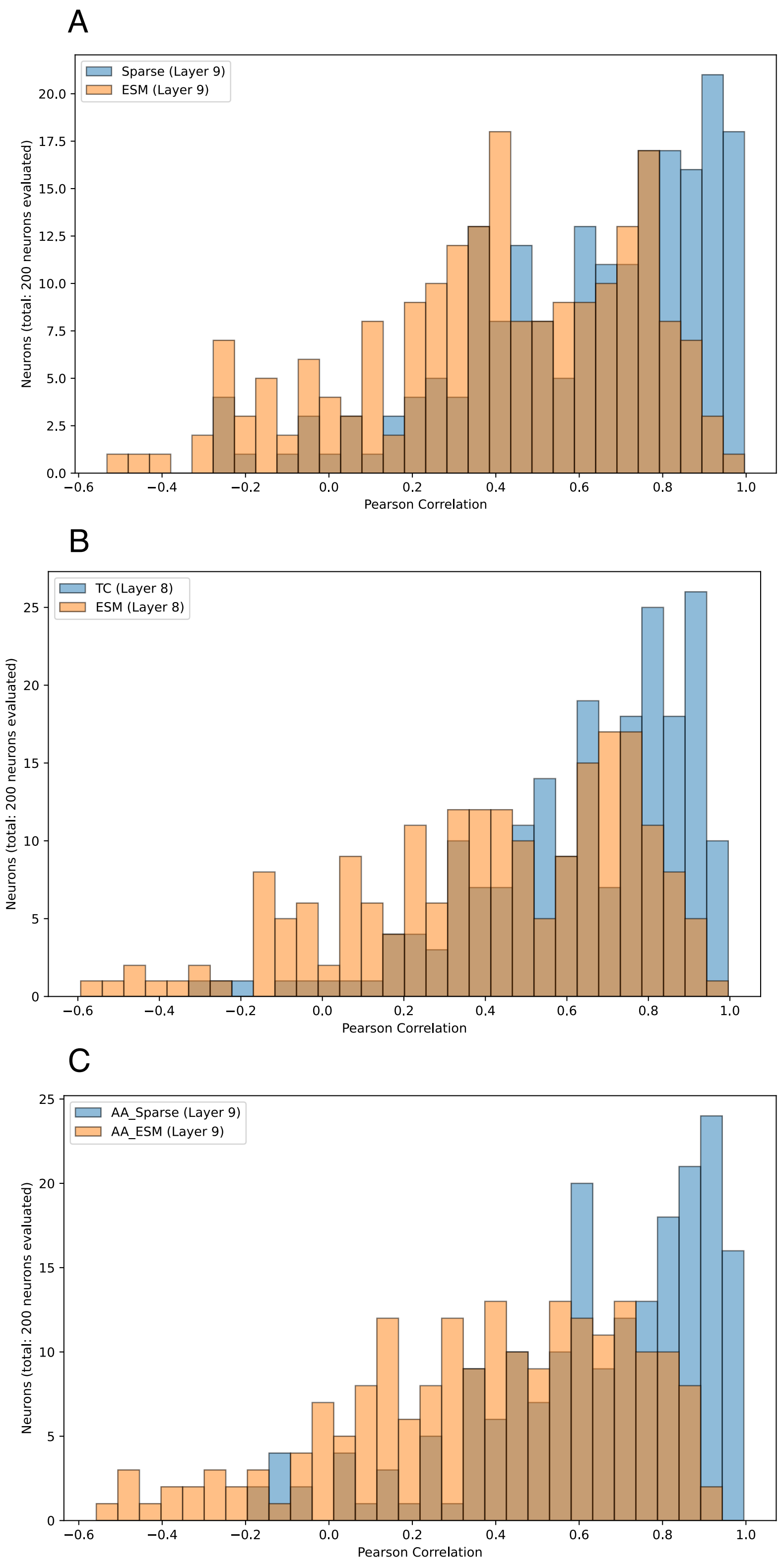

Fig. 3. Overlaid correlation histograms of sparse features and ESM neurons, with the overlaid portion of the histogram shaded in brown (made by using the same range for both histograms): (*A*) Sparse features from SAE (layer 9) and ESM neurons from layer 9—for protein-level representations. (*B*) Sparse features from Transcoder (TC) (layer 8) and ESM neurons from layer 8—for protein-level representations. (*C*) Sparse features from SAE (layer 9) and ESM neurons from layer 9—for AA-level representations. (2 nan values were discarded for the SAE, leaving 198 neurons).

of the multilayer perceptron sublayer of a transformer block at the token level in NLP. But we train our transcoders to learn a sparse approximation of the computation needed to get from the protein-level representation in layer L to the protein-level representation in layer L+1. So the sparse features in our transcoder are features that are involved in transforming one layer's protein-level representation into the next. The sparse features thus extracted are comparably interpretable to those from SAEs as shown in Table 2, with median interpretability scores of 0.66 to 0.73 for the layers examined. Fig. 3*B* is an overlaid histogram showing the interpretability scores for the sparse features from the transcoder trained on layer 8's protein-level representation as well as the interpretability scores of ESM neurons in layer 8's protein-level representation. It shows several highly interpretable sparse features (with Pearson correlations between 0.7 and 1) compared to ESM neurons.

Table 4. Examples of several SAE neurons (from various layers) with very high interpretability and some median interpretability, with Claude's interpretation, for AA-level representations

Model	Neuron no.	Claude's interpretation	Correlation
Sparse (layer 10)	7554	This neuron appears to be detecting proteins involved in olfactory or gustatory sensory perception, particularly insect chemoreceptors and taste receptors.	0.987
Sparse (layer 8)	11965	This neuron appears to be detecting proteins with sulfotransferase activity, particularly those involved in steroid or lipid metabolism.	0.714
Sparse (layer 9)	15537	This neuron appears to be primarily detecting proteins with peptidyl-tRNA hydrolase activity, particularly those belonging to the PTH family and involved in translation processes.	0.995
Sparse (layer 10)	16441	This neuron appears to be detecting proteins that function as transmembrane transporters, particularly those belonging to the Major Facilitator Superfamily or other transporter families.	0.711
Sparse (layer 9)	19717	This neuron appears to be detecting proteins belonging to the UPF0312 family, specifically those located in the periplasmic space.	0.981
Sparse (layer 9)	10718	This neuron appears to be detecting proteins that belong to the Cation transport ATPase (P-type) (TC 3.A.3) family, particularly those involved in transporting metal ions like calcium, potassium, or copper across membranes.	0.974
Sparse (layer 7)	10650	This neuron appears to be primarily detecting proteins with calcium ion binding functionality, often associated with muscle-related or calcium-responsive cellular processes.	0.967
Sparse (layer 9)	1346	This neuron appears to be detecting proteins involved in molybdopterin cofactor biosynthesis or related enzymatic activities, particularly those with molybdopterin molybdotransferase or similar functions.	0.990
Sparse (layer 7)	12791	This neuron appears to be detecting proteins with adenine deaminase activity, particularly those belonging to the Metallo-dependent hydrolases superfamily and Adenine deaminase family.	0.998

The correlation score, ranging between 0 and 1, represents the interpretability of the SAE neuron. See *SI Appendix*, section S5 for full list.

Table 6 presents examples of Claude's automated interpretations for both very highly interpretable (0.9+ scoring) and near-median level interpretable sparse features extracted by transcoders. Many of these features align with specific protein families and functions. For instance, sparse feature 9270 (in layer 9) is associated with proteins that function as arginine-tRNA ligases, while feature 12680 in layer 9 captures proteins involved in apoptosis, especially caspases. Additionally, feature 15097 (layer 8) is associated with proteins from the Short-chain dehydrogenases/reductases (SDR) family, particularly those involved in fatty acid elongation and retinol metabolism. Another sparse feature (4745 in layer 7) captures proteins from the chemokine family, which are involved in immune response and cell signaling processes, while sparse feature 19393 in layer 10 is associated with UDP-glycosyltransferase activity. These findings demonstrate that transcoders are able to extract sparse features that align with biologically meaningful features like protein families and functions in an unsupervised manner, underscoring their potential as an alternative method to SAEs for understanding protein-level PLM representations.

Since SAE feature interpretations tend to correspond to family and function, we expect an increased sequence similarity between coactivating proteins. We performed experiments to verify that this is the case. We first measured both the sequence similarity

Table 5. Comparison of SAE and ESM neurons' autointerpretability scores for various layers for the amino acid-level representations

Model (layer)	Median correlation	Mean correlation	%age with $P < 0.05$	Neurons analyzed	Neurons attempted
Sparse (layer 6)	0.673	0.616	74.5%	200	220/20,000
ESM (layer 6)	0.499	0.441	51.8%	200	202/480
Sparse (layer 7)	0.680	0.593	73.5%	200	208/20,000
ESM (layer 7)	0.442	0.384	44.5%	200	201/480
Sparse (layer 8)	0.703	0.589	71%	200	209/20,000
ESM (layer 8)	0.456	0.423	48%	200	201/480
Sparse (layer 9)	0.714	0.645	75.8%	200	206/20,000
ESM (layer 9)	0.416	0.380	44%	200	201/480
Sparse (layer 10)	0.703	0.625	73.5%	200	208/20,000
ESM (layer 10)	0.438	0.393	45.7%	200	201/480

Sparse neurons are more interpretable than ESM neurons. Here, P-value means P-value of the Pearson correlation (i.e. the autointerpretability score), so that column indicates the percentage of neurons with $P < 0.05$.

Table 6. Examples of Transcoder neurons (from various layers) with very high interpretability and near-median interpretability, with Claude's interpretation, for protein-level representations

Model	Neuron no.	Claude's interpretation	Correlation
TC (layer 9)	9270	This neuron appears to be detecting proteins that function as arginine–tRNA ligases (arginyl-tRNA synthetases) involved in arginyl-tRNA aminoacylation.	0.989
TC (layer 9)	12680	This neuron appears to be detecting proteins involved in apoptosis, particularly caspases and related proteins that play a role in programmed cell death pathways.	0.680
TC (layer 8)	12186	This neuron appears to be detecting proteins with FMN-dependent NADH:quinone oxidoreductase activity, particularly those in the Azoreductase type 1 family.	0.937
TC (layer 8)	12187	This neuron appears to be detecting proteins involved in metal ion binding and transport, particularly those related to iron, copper, and other metals.	0.711
TC (layer 8)	15097	This neuron appears to be detecting proteins belonging to the Short-chain dehydrogenases/reductases (SDR) family, particularly those involved in fatty acid elongation or retinol metabolism.	0.747
TC (layer 7)	4745	This neuron appears to be detecting proteins belonging to the chemokine family, particularly those involved in immune response and cell signaling processes.	0.989
TC (layer 10)	19393	This neuron appears to be detecting proteins with UDP-glycosyltransferase or UDP-glucuronosyltransferase activity, primarily from the UDP-glycosyltransferase family.	0.943
TC (layer 10)	4745	This neuron appears to be detecting proteins involved in GTP 3′,8-cyclase activity and molybdenum cofactor biosynthesis, particularly those belonging to the MoaA family within the Radical SAM superfamily.	0.673

The correlation score, ranging between 0 and 1, represents the interpretability of the Transcoder neuron. See *SI Appendix*, section S5 for full list.

between the set of coactivating proteins and the sequence similarity between the set of non-coactivating proteins in the set of interpretation sequences for a neuron (i.e. SAE feature) analyzed in (for example) layer 9's AA-level SAE. We find that the coactivating proteins have increased sequence similarity as measured by global alignment, local alignment, and k-mer ($k = 3$) scores compared to non-coactivating proteins (−0.5304, 0.4272, 0.0558) vs. (−1.7535, 0.2149, 0.0287). However, the Pearson correlation between the sequence similarity scores for a neuron and the neuron's interpretability score is weak (0.263, 0.117, 0.188)—i.e. our SAE neurons with greater sequence similarity among their coactivating proteins are not necessarily more interpretable—suggesting that there is likely more to an interpretable SAE neuron than increased sequence similarity. Next, for layer 9's protein-level SAE, the coactivating proteins have increased sequence similarity compared to non-coactivating proteins (−0.8310, 0.6009, 0.0528) vs. (−1.8392, 0.2276, 0.0276), and the Pearson correlation between the sequence similarity scores for a neuron and the neuron's interpretability score is also weak (0.200, 0.135, 0.142). *SI Appendix*, Table S5 provides this information for all other layers.

SI Appendix, section S2 contains some additional tables related to our results. *SI Appendix*, Table S1 shows the percentage of interpretable features in our SAEs and transcoders at various interpretability thresholds. For instance, for layer 9's AA-level SAE: If we employ a threshold of 0.7 for a neuron being considered interpretable, then 50.50% of the randomly analyzed neurons are interpretable; it is 74.00% (threshold 0.5), 62.50% (0.6), 37.50% (0.8) at some other thresholds. In *SI Appendix*, Table S2, we indicate the level of sparsity for neurons in the various SAEs and transcoders at all layers, by providing the mean and median percentages of the number of proteins that activate a sparse neuron. For instance, for the AA-level SAE (layer 9), the mean activation percentage of a neuron is 2.36% of the proteins and the median activation percentage of a neuron is 0.65% of the proteins. While for the protein-level SAE (layer 9), the mean activation percentage of a neuron is 0.32% of the proteins and the median activation percentage of a neuron is 0.02% of the proteins. Interpretations for all sparse features analyzed (across all trained SAEs and transcoders) and ESM neurons, along with their interpretability scores, are available in *SI Appendix*. See *SI Appendix*, section S5 for more details.

Discussion

Our work extracts interpretable features in multiple ways from protein-level representations using sparse autoencoders and transcoders. Both kinds of representations, derived from PLMs, benefit from interpretability. Protein-level representations are widely used in protein-level downstream tasks, making their interpretability crucial for human–AI collaboration and enhancing trust in downstream bio-AI applications. Extracting interpretable features concealed in a particular representation can help researchers explain and debug downstream performance, and optimize the choice of representation layer for different downstream applications.

On the other hand, interpreting AA-level representations sheds light on the biological features captured by PLMs at each layer. In SAEs trained on both kinds of representations, we identify sparse features that correspond to specific gene ontology terms, protein families, and protein functions. SAEs and transcoders disentangle these features in a completely unsupervised manner, thus providing a means to uncover biologically relevant information. These sparse features could potentially be used to learn more about both well-known and poorly understood protein families and functions, and about how they interact.

Our work highlights both SAEs and transcoders as promising unsupervised approaches for disentangling interpretable features in protein-level representations. As PLMs continue to get larger and better, we expect SAEs and their variants to play a more integral role in uncovering interpretable features present within their representations.

As open-source PLMs become more powerful, it is also vital to evaluate whether they unintentionally encode sensitive or